When Instance Normalization Hides the Level: An Audit of Exogenous Covariate Forecasting

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Abstract

Deep forecasters ship instance normalization (RevIN and its successors) as a default: each input window is standardized by its own statistics and the removed level restored at the output, so the level a forecast returns to is guessed from the target's own recent past. In energy forecasting that guess is needless — the level is set by observable weather, and at the wind farms we audit the archived forecast pins it more than four times better than the window mean does. We toggle the layer under information parity, with identical covariates in every arm, and the audit does not return a cost: the penalty's size and even its sign flip with how the covariates are scaled against the window-normalized target, a convention covariate forecasters fix without discussion. What survives every such convention is a boundary — the level the layer discards has to be rebuilt from the covariates, and only two of the five backbones in our parity grid reliably manage it. We propose no new normalizer, and we raise this as an open problem: our results expose a level-recovery boundary that existing covariate forecasters should explicitly test, and the field currently reports normalization results without reporting the convention that decides their sign. A cheap pre-training score, described in the body, tells us in advance which side of the boundary a dataset falls on.

1 The problem

Renewable generation and electricity demand have levels set by observable physical drivers — wind by the weather regime, solar by cloud cover, load by temperature — and forecast in advance by numerical weather prediction (NWP). Conditioning on that level is standard practice, from the GEFCom competitions to operational wind and load models [1, 2].

Modern deep forecasters lead instead with instance normalization (RevIN) [3]; successors refine *which* endogenous statistics to use [4–6], but all read them off the target's own recent past (Figure 1). The default has been carried into covariate forecasters without a reported test: CrossLinear and TimeXer both normalize by default and neither ablates the choice [7, 8]. Berthelier et al. [9] find RevIN "discards potentially predictive context", but in an explicitly univariate setting, and defer exogenous covariates while "anticipat[ing] similar conclusions"; pre-training selectors read endogenous statistics instead [10, 11]. We know of no published table that toggles the layer with informative covariates held identical across arms.

This paper is that table, and what it returns is not the number we expected. Under parity the layer's cost is not portable: its size and sign flip with the covariate *footing* — how the covariates are scaled against the window-normalized target — through a one-line mechanism (§3). What no footing

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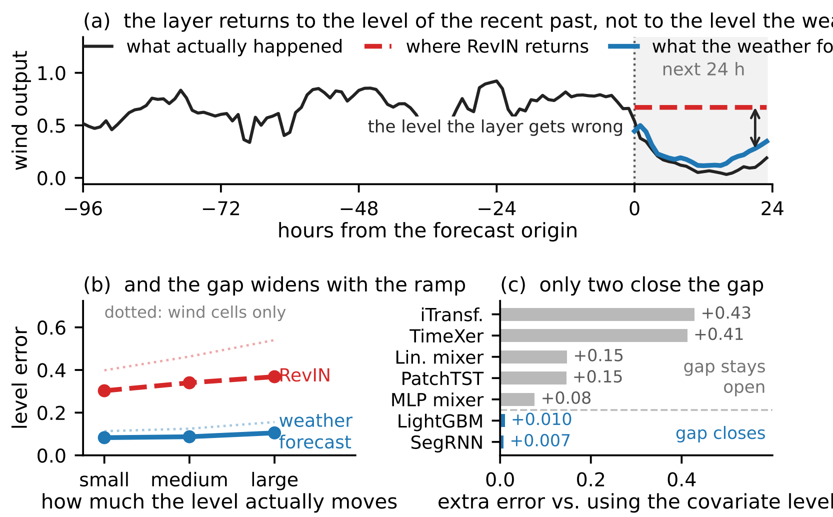

Figure 1: **The level a forecaster restores, against the level the weather forecast implies. (a)** One origin, chosen by rule (Appendix A): RevIN restores a level far from both the realized one and the NWP-implied one. Over all 3,315 test origins the mean absolute level errors are 0.178 and 0.042. **(b)** The same two rules over every origin of the seven clean cells, by tercile of realized in-horizon ramp, train-sd units; the covariate-implied level is CN's train-fitted first stage, with no backbone, seed or footing in it. **(c)** With the layer on, the seed-averaged gap to CN over the eleven exogenous cells, the two with disclosed provenance defects included (§3): the five parity backbones plus two replicated architectures. SegRNN and LightGBM (blue) close the gap from the same covariates; on this implementation the other five do not.

changes is a boundary: the level the layer discards must be rebuilt from the covariates, and only two of the five backbones in the parity grid reliably close the resulting gap. A score computable before training calls the winning side on all eight datasets (§2). We bring this to the workshop as an open problem rather than a solution, and Appendix A carries the full accounting.

2 A boundary, and a score you can compute first

We compare three ways of handling the level: global z-scoring (RAW), instance normalization (IN), and covariate-conditional normalization (CN), which subtracts a train-fitted covariate-predicted level, models the residual and adds it back. CN is an instrument, not a proposal — a plain first-stage regression matches it — and what we endorse is conditioning the level on covariates already in hand. With $\bar{y}$ and s the lookback window's mean and standard deviation, any IN-equipped backbone is

$$\hat{y} = \bar{y} + s\, f\big((x - \bar{y}\mathbf{1})/s,\ g\big) \tag{1}$$

for any covariates g. The backbone predicts a *shape*; the level is restored afterwards, and f never sees $\bar{y}$, so any correction of the window's level has to come from the covariates alone. That is the whole boundary, and §3 shows where backbones fall on either side of it.

IN pays for within-horizon level drift, CN for the level variance the covariates fail to explain, so CN wins once the covariate-explained share of the level passes a crossover that falls with the horizon (Appendix B). That share is unobservable. Our proxy for it, on a different scale and not an estimator of it, is the **Level Predictability Score**: the out-of-sample R^2 of window-mean levels on window-mean covariates under expanding chronological cross-validation (Eq. 2). It needs no training run. The rule is one line — LPS $\geq \tau \Rightarrow$ do not set the level endogenously, otherwise instance normalization — with τ=0.3 fixed before any run of the grid.

What the score does not establish. Classification is easy on our panel, and τ is not what makes it so: the values are bimodal (energy 0.744–0.894, benchmarks −0.717 to 0.283) and any $\tau \in$

$[0.30, 0.70]$ separates them identically. Bimodality is not a validation of the threshold. Its lower end sits 0.017 above Electricity, and at its upper end five of the ten wind zones we disaggregate fall below τ while covariate-conditioned handling wins every one of them — so the agreement is a statement about the eight datasets alone. The panel's top is soft the other way: GEFCom-Load's 0.894, the highest we report, is read off realized rather than forecast temperature, so it is optimistic and widens the gap. And the score's evaluation blocks span the full series, test segment included — computable before training is not computable before the forecast origin. Confining the blocks to pre-origin data leaves every side intact — $8/8$ datasets, $10/10$ zones — but not every margin: **Electricity moves to 0.2997, three ten-thousandths below** τ, and the benchmark values swing hard in both directions (Weather $+0.110 \rightarrow -1.347$, ETTh2 $-0.205 \rightarrow +0.188$). What the choice of evaluation block decides is those values, not their side (Appendix A).

3 What the toggle returns

Setup. The grid crosses five normalization arms {RAW, RevIN, SAN, FAN, CN} [3–5] with four backbones {RLinear, PatchTST, SegRNN, LightGBM} [12–15], three horizons and five seeds, over four exogenously driven energy series (Jeju Wind, self-curated; GEFCom-Wind/-Load/-Solar [1]) and four standard benchmarks. Wind and solar covariates are archived operational forecasts, never reanalysis; splits are chronological; lookback is frozen, so arms differ only in normalization. Two of the eleven exogenous cells carry a disclosed provenance defect — GEFCom-Load, whose covariate is realized rather than forecast temperature, and Jeju h=48, whose implementation consumes the $h \leq 24$ forecast band — so every count below is given on both bases. Grid, accounting, intervals and reproducibility limits: Appendix A.

The cost is not a portable number. Under information parity — identical covariates in every arm, only the layer toggled — instance normalization costs $+0.016$ MSE pooled, and $+0.020$ on the one bit-reproducible backbone.[1] Then the objection: covariates are scaled globally in every arm while RevIN rescales only the target, so their relative scale moves window to window. Varying only that *footing* (Table 1) moves the toggle to $+0.026$, -0.075 and -0.030 across three of the four conventions we try. At its most favourable footing the layer is simply not worse. We therefore withdraw the penalty as a portable number and report what generates it.

The mechanism: the window scale leaks into the covariate slope. Instance normalization multiplies the backbone's output by s, so a covariate term βg formed inside f reaches the forecast as $s\beta g$: the effective covariate slope varies with the target's own volatility. Dividing the covariates by the same s restores it, since $s\beta(g/s) = \beta g$ — a repair of the footing, not a proposal. It replicates on a nonlinear backbone, where the identity is not exact.

What survives every footing. Give each cell its *best* endogenous configuration anywhere in the grid and covariate-conditioned level handling still wins every one of them. What the gap depends on is the backbone, not linearity: the transformers sit far from CN while SegRNN and LightGBM, which close it from the same covariates, sit next to it (Fig. 1(c)). We report this as a property of the architectures *as we configured them*, not as an impossibility. The boundary is about what a backbone does with the covariate level, not about access to it, and nothing endogenous closes the gap either. Margins, the per-backbone ladder and the isolated toggle on TimeXer and iTransformer — mixed in sign, so the layer is not shown to harm those two — are in Appendix A.

The score, and the failure side it guards. The LPS values and the predicted gap signs were committed before the first run. All eight signs were confirmed, unchanged under MSE, MAE and MASE [16] (Fig. 4) — but the eight datasets form two correlated families, energy and standard

[1] Three of the five backbones the pooled value covers carry dropout, whose CPU masks are thread-count dependent, so that value is conditional on the original run's thread configuration. The linear mixer is not, and is printed beside it at every mention.

benchmarks, so this is per-dataset evidence rather than eight successes, and it is the full-series variant of the score that makes the calls. The rule is one-sided because the low-LPS side fails as the score anticipates: CN reaches 6.60 MSE on ETTh2 (Table 2) by overfitting a calendar, which LPS flags at -0.205 before any training.

4 What we are asking the workshop

Our results expose a **level-recovery boundary that existing covariate forecasters should explicitly test**: whether the backbone, with the layer on, can rebuild from the covariates the level the layer removed. In the primary parity grid two of five backbones reliably close that gap; the same boundary appears in two further architectures we replicate, although their isolated normalization toggles are mixed in sign. One rule of practice follows immediately — report the covariate scaling convention beside any instance-normalization result, since on our grid that convention decides the sign of the effect and none of the papers we checked states it — and where the score clears, condition the level on covariates already in hand; a first-stage regression suffices.

We give no megawatt or emissions figure. Reserve is sized on an upper quantile rather than a mean, reserve capacity is not displaced energy [17, 18], and unit commitment is a threshold response with no smooth map from our MSE. What we can report is what a reserve decision consumes: after identical split-conformal recalibration [19] of every arm, the day-ahead 80% interval is 23 to 50% narrower under covariate-conditioned level handling, at equal or better coverage on three of four datasets, while the reserve-relevant upper tail stays loose for both.

Limitations. Two backbones and two GEFCom sets in the footing grid; one split; the Jeju band defect (§3); all evidence is energy series, the mechanism and the score being domain-agnostic in form but untested elsewhere; Fig. 1(c)'s architecture dependence is measured, not explained; and the theory covers the linear case. Every number above is re-derivable from frozen CSVs by a script in the released repository, and Appendix A carries the accounting, every pre-registered endpoint that did not confirm, and a grading of what "pre-registered" is worth block by block.

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A Setup details, accounting, and availability

A.1 Data, grid and accounting

Data and splits. Hourly resolution throughout. Exogenous group: Jeju Wind (self-curated; development dataset) and GEFCom2014 Wind/Load/Solar [1] (competition-provided target-time forecast covariates for wind and solar; realized station temperature for load). Jeju's covariates are archived KMA wind-speed forecasts averaged over nine wind-farm grid cells. The archive contains a single daily issue, 23:00, at leads 26–73 h, so the $h{=}24$ band is the 23:00 issue of $D{-}2$ (leads 26–49 h) and the $h{=}48$ band that of $D{-}3$ (leads 50–73 h): both are settled well before day-ahead gate closure,

Table 1: **The footing grid: the audited penalty is not a portable number; the linear mixer's boundary is.** Linear mixer, seed-averaged test MSE in train-fit global-z units. Each column pair runs the same layer toggle and changes only the covariate footing (§1). The Cn column is the frozen parity block's, comparable because the linear mixer is bit-reproducible; † variance-floored. The body reads this table in §3.

	global z		window†		scale-matched		
Cell	Raw	RevIN	Raw	RevIN	Raw	RevIN	Cn
GEFCom-Wind h=24	0.248	0.320	0.287	0.229	0.271	0.247	0.177
GEFCom-Wind h=96	0.250	0.307	0.351	0.251	0.287	0.254	0.167
GEFCom-Wind h=336	0.381	0.419	0.441	0.364	0.374	0.381	0.199
GEFCom-Solar h=24	0.161	0.146	0.155	0.138	0.169	0.147	0.064
GEFCom-Solar h=96	0.146	0.146	0.208	0.181	0.171	0.135	0.061
GEFCom-Solar h=336	0.192	0.195	0.615	0.446	0.251	0.178	0.066
mean	0.230	0.255	0.343	0.268	0.254	0.224	0.122

Table 2: **Context, not contribution: access versus layer.** Test MSE (train-fit global z), main grid, averaged over backbones, horizons and seeds; SAN/FAN exclude LightGBM; **bold** = best in the row. Only Cn consults covariates here, so these gaps combine access with the layer and locate the boundary rather than isolate the layer. Between Raw and the best endogenous arm, neither of which sees covariates, the layer helps on GEFCom-Wind and -Load, which is why the parity design and not this table is the relevant comparison. †reference cell (realized, not forecast, temperature).

Dataset	Raw	RevIN	SAN	FAN	Cn
Exogenously driven					
Jeju Wind	0.6887	0.6969	0.7030	0.7724	**0.2933**
GEFCom-Wind	1.1153	1.0039	0.9755	1.1875	**0.1629**
GEFCom-Solar	0.1828	0.1844	0.2116	0.1833	**0.0580**
GEFCom-Load†	0.3592	0.3432	0.3330	0.3678	**0.1066**
Standard LTSF					
ETTh1	0.3962	**0.3771**	0.3910	0.4114	2.2016
ETTh2	0.3868	0.2900	**0.2869**	0.3253	6.6044
Electricity	0.1500	0.1458	0.1443	**0.1375**	0.1726
Weather	0.1815	0.1719	**0.1666**	0.1747	0.7274

which is the property §2 needs. That is the band assignment the design intends; the implementation fits one first stage per dataset, outside the horizon loop, so the h=48 cells in fact consume the $h \leq 24$ band—the defect Limitations reports. GEFCom2014 publishes no issue timestamp for its forecast covariates, so their availability rests on the competition shipping the whole target-month block in advance rather than on a lead we can verify; we disclose that rather than assert a lead. Standard group: ETTh1, ETTh2 (electricity-transformer temperature), Electricity, Weather under literature splits (ETT 12/4/4 months; 7:1:2 elsewhere); zone series use chronological 60/20/20. Global z-scoring is fit on the train segment only; splits are strictly out-of-time.

Run accounting. Main grid $1{,}794$ runs: 5 arms $\times$ 3 torch backbones $\times$ 23 $(\text{dataset}, h)$ cells $\times$ 5 seeds, plus 3 applicable arms $\times$ 23 cells for LightGBM, which is deterministic and run at one seed. Horizons are $\{24, 96, 336\}$, $\{24, 48\}$ on Jeju Wind by forecast-band coverage, hence *eleven* exogenous cells. One convention, so that no base has to be guessed: "eleven exogenous cells" always includes the two carrying disclosed provenance defects (GEFCom-Load, Jeju h=48); where the clean-seven restriction was also computed it is printed beside the eleven-cell figure, and where no such figure appears it was not computed rather than omitted. The restriction is computed for the parity block's per-backbone endpoints, the ramp split and the fed-back block; it is not computed for the SOTA replication, the per-backbone Raw-to-Cn gaps or the pinball comparison, which are eleven-cell figures only. LightGBM$\times$\{SAN, FAN\} is structurally inapplicable (torch statistics predictors).

Information-parity block, $1{,}133$ runs over five covariate-capable backbones: LightGBM-Cov, linear mixer, MLP mixer, SegRNN-Cov, PatchTST-Cov, every arm given identical past and future covariates. *SOTA replication*, $1{,}275$ runs (TimeXer and iTransformer, each in a covariate-fed and an

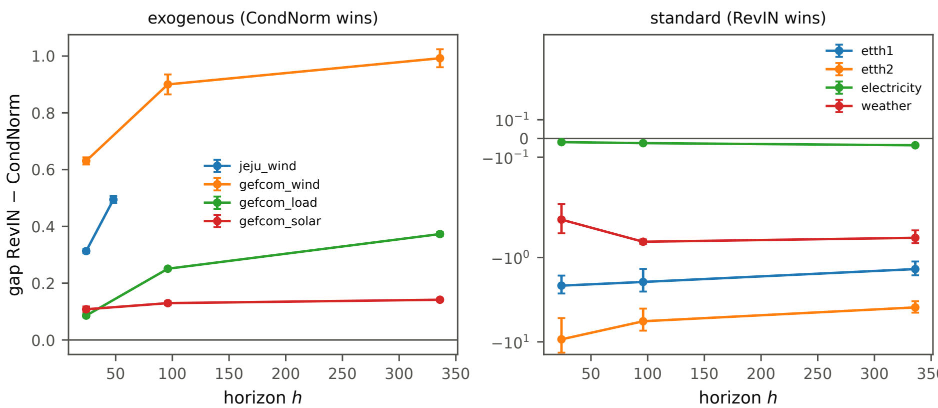

Figure 2: **The Table 2 dissociation, by horizon.** Main-grid RevIN−Cn gap per (dataset, h), mean ± 1.96 SEM over backbones and seeds. On the exogenous group the gap is positive and grows with the horizon — the direction Appendix B predicts for a frozen window statistic — while the standard group reverses (symlog axis). Colours index datasets within each panel (separate legends); Jeju Wind appears only on the left and stops at h=48, because its archived forecast bands cover only $h \in \{24, 48\}$ (Data and splits below). Like Table 2, these arms combine covariate access with the layer; the isolated toggle is the parity block below.

endogenous variant, built-in normalization removed and arms injected; the count is uneven because two model-variant/arm combinations were run at fewer horizons): Cn is best on all four exogenous datasets in both, mean RevIN−Cn gap $+0.41$, while the isolated Raw-versus-RevIN toggle is mixed in sign: over the eleven exogenous cells it is $+0.011$ on TimeXer and $+0.019$ on iTransformer, positive in 5 of 11 each, ranging -0.085 to $+0.135$ and -0.061 to $+0.152$. We give these rather than the aggregate alone because these are the models §1 names, and on them the layer is not shown to be worse. Total 4,202. The graded zone study adds 450 pre-registered RLinear runs (10 zones $\times$ 3 arms $\times$ 3 horizons $\times$ 5 seeds); a deferred LightGBM arm is incomplete and excluded from the primary endpoint, disclosed at pre-registration analysis time. The footing block adds 480 runs at a pinned thread count: 2 arms $\times$ 4 footings $\times$ 3 horizons $\times$ 5 seeds on each of the two GEFCom sets (240) for each of the linear and MLP mixers, of which the linear-mixer half is Table 1. The two level-isolating footings add 120 and the fed-back-statistics arm 70, both on the linear mixer. The mean-only ablation is 180 runs and the ramp split trains nothing. The 2,640-run synthetic grid behind Appendix B is a separate tally, outside the 4,202.

Compute. Summing the logged wall-clock over every row of the frozen main-grid file (main grid plus SOTA replications) gives 179 hours on one workstation—a single consumer GPU plus CPU LightGBM, which accounts for 42 of those hours. The remaining blocks were not wall-clock logged at the time and are not included in that total, which therefore covers about half the runs behind this paper; the footing, level-isolating and fed-back blocks together are a few hours of the same workstation. Deduplicating the 48 superseded re-run rows leaves 154 hours for the 3,069 runs actually reported; the information-parity block's wall-clock was not logged. The LPS screen itself was not instrumented at pre-registration time; re-measured on one pinned core it takes 0.036 s per univariate series (median of 7, GEFCom-Wind; 0.048 s including load and featurization). The full eight-dataset panel, including every channel of the multivariate benchmarks, completes in minutes.

A.2 The information-parity block

The parity block, and how far it travels. This is the block the body withdraws as a portable number; we report it in full because the footing result is a statement *about* it (sample forecasts: Fig. 3). Per-backbone isolated cost at the Global footing, eleven cells: MLP mixer $+0.024$ $[+0.012, +0.039]$, linear mixer $+0.020$ $[+0.006, +0.037]$, LightGBM-Cov $+0.019$ $[+0.012, +0.027]$, SegRNN-Cov $+0.011$ $[+0.006, +0.017]$, PatchTST-Cov $+0.004$ $[-0.009, +0.017]$ (null); as a share of each backbone's own Raw+cov error, 12.1, 7.4, 15.0, 8.5 and 1.3 per cent. Pooled over backbones and cells the toggle costs $+0.016$ for RevIN (against

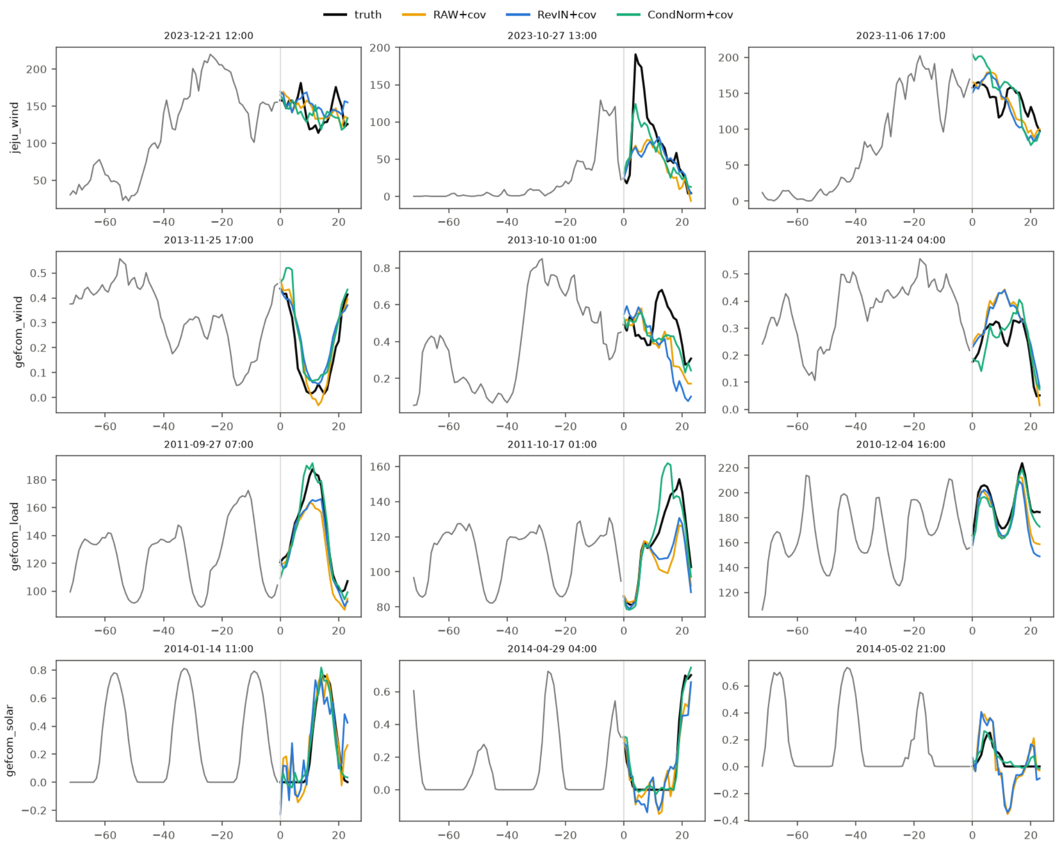

Figure 3: **What the parity arms actually predict.** Test-segment forecasts (h=24, lookback 336) from the three parity arms of a linear mixer given identical exogenous covariates — Raw+cov, RevIN+cov, Cn+cov — on the four exogenous datasets (rows), three origins each; grey is the lookback, black the realized target. Retrained at one seed for illustration; no number in the paper is read from this figure. Note RevIN+cov re-entering at the lookback's level where the realized series has moved (wind rows).

$+0.020$ for the dropout-free linear mixer alone, just above), $+0.048$ for SAN and $+0.154$ for FAN, every interval excluding zero—at this footing the successors that refine *which* endogenous statistic to use lose more, not less. All three pooled values are conditional on the thread configuration of the original run, since three of the five backbones they pool carry dropout (see the reproducibility paragraph below); the linear-mixer column is not, which is why the body prints it beside the pooled figure and never quotes the pooled figure on its own. Per-backbone gaps from Raw to Cn over the same eleven cells: linear mixer $+0.127$, MLP mixer $+0.052$, PatchTST-Cov $+0.142$, SegRNN-Cov -0.004, LightGBM-Cov -0.009; from RevIN to Cn, $+0.147$, $+0.076$, $+0.145$, $+0.007$ and $+0.010$ respectively. On the six footing-grid cells, LightGBM-Cov's Raw arm matches or beats Cn on four of six. These are the numbers behind the body's scoping of the boundary: SegRNN-Cov and LightGBM-Cov close the gap from identical covariates endogenously, so Cn is evidence about level handling, not a method recommendation. The boundary replicates on TimeXer and iTransformer, though the isolated toggle there is mixed in sign and its numbers are above [8, 20]. A conjecture we record but have not tested, on why the boundary follows architecture: under the layer, using a covariate level requires the backbone to synthesize from its inputs the window-dependent rescaling that Eq. 1 imposes; the two backbones that close the gap route covariates through stateful or partition-based nonlinearities (recurrent gating, tree splits), while the fixed feed-forward and attention maps here do not close it despite being nonlinear.

How the intervals are computed. The inference resamples the eleven seed-averaged (dataset, h) cells i.i.d. with replacement (B=10^5, percentile, fixed seed); Monte-Carlo jitter over B and seed is ≤ 0.0004 on the eleven-cell endpoints; on the six- and seven-cell intervals the percentile bounds move by up to 0.002 across bootstrap paths, so the final digit of those bounds is path-dependent. Three decimals are quoted throughout for comparability. Cells are seed means for the four torch backbones; LightGBM-Cov is deterministic, so its interval carries cell variation only. Across all 55 (backbone, cell) pairs the isolated layer cost ranges -0.026 to $+0.085$ and is positive in 46. Three panels, reported because they do not agree in the same place. (i) Eleven cells, as printed: four of five CIs exclude zero, PatchTST-Cov null. (ii) Resampling whole *datasets* (four clusters) instead of cells: means unchanged, the linear mixer widens to $[-0.001, +0.046]$, so three of five exclude zero. (iii) Eight cells, dropping the GEFCom-Load reference cell: all five exclude zero, because the three Load cells are PatchTST-Cov's three largest negatives ($-0.026/-0.022/-0.024$ at h=24/96/336) and only one negative survives their removal (GEFCom-Wind h=336, -0.018), giving $+0.014$, $[+0.002, +0.025]$, 7/8 positive; on the strictest seven-cell base, which also drops Jeju h=48, PatchTST-Cov returns to null ($+0.012$, $[-0.001, +0.024]$) and the linear mixer sits on the boundary. What survives every panel is the sign: all five backbone means are positive ($+0.004$–$+0.024$ on eleven cells, $+0.012$–$+0.022$ on the strictest base).

What "the layer" means in this block. Toggling RevIN removes the window mean, the window scale and its learnable affine together, so this block measures the layer *as shipped*. The mean-only ablation below separates those channels.

Zone-study secondaries, the Model Confidence Set, and multiplicity. the endogenous RevIN−RAW per-zone prediction is not confirmed (positive in 4 of 10 zones; Spearman -0.48, p=0.16)—the layer's small cost, of parity-block magnitude, sits below zone-level noise. The Model Confidence Set [21] is computed at $\alpha = 0.10$ per (dataset, h) on backbone-averaged squared-error series; all eleven exogenous cells retain CN alone (eight of eight without Load). The MCS is a singleton in 20 of 23 cells on this single evaluation path, so we read it as a compact ranking consistent with the cellwise comparisons above, not an uncertainty set. The five backbone intervals and the pooled RevIN/SAN/FAN ones are per-comparison and uncorrected, and they share the same eleven cells, so "four of five" is replication across backbones rather than a family-wise statement; under the dataset-clustered panel it is three of five, which is why we report the range.

What could replace CN. Replacement costs, all within Block E so the comparison is internal (test MSE, jeju/wind/load/solar): first-stage regression $0.29/0.16/0.14/0.06$, CN $0.31/0.18/0.11/0.06$, ridge dynamic regression $0.29/0.17/0.37/0.14$. Above the threshold the covariate-conditioned replacements sit within hundredths of one another on the wind sets, which is why we treat CN as an instrument rather than a proposal. On the Jeju h=24 cell the ridge dynamic regression is in fact the best of them—$\mathrm{nMAE} = 0.0899$ against CN's 0.1001, a 14.6% cut where CN gives 5.0%—though it is one deterministic run and it fails where the temperature–level response is nonlinear (GEFCom-Load 0.37 against CN's 0.11). We report it because the operational number belongs to the level-conditioning practice, not to our instrument.

Two further checks on the parity block. All standard-group loss series are segment-aware (no windows across curated gaps). Single-origin caveat: promoting the validation segment to a second chronological origin reproduces all four exogenous signs, so the headline dissociation is not an artifact of the one test path.

Which channel of the layer carries the parity-block cost. A mean-only arm, recorded in advance of the run but not timestamped (Table 3), separates the three things the RevIN toggle moves at once. Like the parity block itself, it is measured at the GLOBAL footing and therefore inherits the footing caveat above; it is reported here rather than in the body for that reason. Arms {RAW, mean-only, RevIN} on GEFCom-Wind and GEFCom-Solar (third-party, forecast covariates, so neither the Load reference cell nor Jeju's caveats apply), backbones {linear mixer, MLP mixer}, $h \in \{24, 96, 336\}$, five seeds: 180 runs. The mean-only arm subtracts and restores the lookback window mean and nothing else—no instance scale, no learnable affine—so against RAW it differs in exactly one channel. Datasets, backbones, endpoints and the meaning of each outcome were fixed in writing before the first run; that record is in the released repository but, unlike the main grid and the zone study, it is not git-timestamped ahead of the run, and we do not claim it is (Table 3). Everything else is the parity block: same runner, same frozen lookback per (dataset, backbone, h), same splits, optimizer, early stopping and loss scale. Decomposition per backbone (equal weight over six cells, percentile bootstrap): linear mixer, mean $+0.0104\,[+0.0043, +0.0170]$ with $6/6$ cells positive, scale+affine $+0.0152\,[-0.0044, +0.0377]$ with $3/6$; MLP mixer, mean $+0.0061\,[+0.0018, +0.0107]$ with $5/6$, scale+affine $+0.0226\,[+0.0057, +0.0463]$ with $5/6$. So the mean channel is the smaller but sign-stable half, the scale channel the larger and sign-unstable one; on the linear mixer's GEFCom-Solar cells the instance scale in fact *helps* at every horizon (-0.016 to -0.003) while the mean channel still costs. By horizon the two separate cleanly on the linear mixer: mean $+0.008/+0.009/+0.015$ at h=24/96/336, scale+affine $+0.021/+0.019/+0.006$. Recorded outcome: PRIMARY 1 (mean channel positive on both backbones) confirmed; PRIMARY 2 (mean carries $\geq 50\%$) not confirmed at 41% and 21%; the horizon secondary confirmed on the linear mixer.

A reproducibility limit, and what we did about it. PyTorch samples dropout masks per intra-op thread chunk on CPU, so any backbone with dropout is reproducible only at a fixed thread count. Re-running the parity block's RAW and RevIN cells reproduces the linear mixer *bit-exactly* and the MLP mixer not at all: varying threads from 1 to 28 moves a single MLP-mixer cell by up to 0.008 MSE, the same order as the effect being measured. Every block added after that discovery pins the thread count, and every primary endpoint is carried by the linear mixer, which has no dropout. The frozen parity block predates the fix, so its three dropout-bearing backbones keep intervals conditional on the thread configuration of the original run; they no longer carry a headline.

A.3 Footing, mechanism and level isolation

The footing block (Table 1), and the endpoint it failed. Recorded before this block's grid, including the withdrawal clauses, but—unlike the main grid and the zone study—written after the main grid had run and therefore not timestamped ahead of it (Table 3). The series is globally z-scored on train statistics before any arm sees it; let s be the per-window standard deviation of the target lookback in those units, the divisor RevIN uses (up to its stabilizing ϵ). The body's toggle values with their intervals: $+0.026$ $[+0.002, +0.052]$ at GLOBAL, -0.075 $[-0.118, -0.037]$ at floored WINDOW, -0.030 $[-0.050, -0.012]$ at SCALE. Four covariate footings: GLOBAL cov_z (level kept, global units—the parity block); WINDOW $(\mathrm{cov}-\bar{\mathrm{cov}}_w)/\mathrm{sd}_w$ (level destroyed, own units); the same with sd_w floored at 0.1, because unfloored it divides by ≈ 0 on all-night solar windows and the arm explodes (1.8, 23.4, 788 MSE); and SCALE cov_z/s (level kept, the target's units). Arms $\{\text{RAW}, \text{RevIN}\}$ are crossed with all four footings, $h \in \{24, 96, 336\}$, five seeds, at a pinned thread count; the CN column of Table 1 is the frozen parity block's, since CN is a reference here rather than a crossed arm. The GLOBAL cells reproduce that block, which is the implementation check. The mechanism decomposition the body points here for: on the linear mixer the RevIN arm's own footing sensitivity is $+0.032$ $[+0.013, +0.054]$, replicating on the nonlinear MLP mixer ($+0.024$, $6/6$ cells positive) where the identity is not exact; of the full $+0.056$ interaction, -0.024 is the RAW arm getting *worse* under a rescaling it has nothing to cancel, so $+0.032$ is the mechanism estimate (exploratory — the six cells span two datasets $\times$ three horizons, so no dataset-level inference).

An earlier control of ours window-normalized the covariates and reported that it "failed to close" the confound. That ruling was wrong and we correct it here: that control changes the units *and* the information at once—recentring each covariate window on its own mean destroys exactly the level at issue—so it could adjudicate neither. SCALE is the units-only version. A same-day amendment to that record, made in writing rather than silently, replaced our first endpoint ($\text{RevIN}_S - \text{RAW}_S$) with a strictly harder one after a smoke run showed RAW_S is not a control but a distortion of the RAW arm, which carries no per-instance scaling for its covariates to match: the endpoint became $\min_f \text{RevIN}_f - \text{RAW}_{\text{GLOBAL}}$, handing the layer its best of four footings against the arm's native one. Both are reported, and both fail: the original endpoint at -0.030, the amended one at -0.014, and the level-preserving-footings variant at -0.006 $[-0.012, -0.001]$. That is what the body withdraws on.

The level-isolating block, in full. Per cell, MSE change from removing the covariate level with the divisor held fixed (linear mixer, 120 runs): RAW at global $\rightarrow$ centred-global, GEFCom-Wind $+0.013/+0.026/+0.008$ and GEFCom-Solar $-0.013/-0.011/-0.017$ at h=24/96/336; RevIN at scale-matched $\rightarrow$ centred-scale, $+0.004/-0.006/-0.034$ and $-0.013/-0.009/-0.019$. Pooled: $+0.001$ $[-0.011, +0.014]$ and -0.013 $[-0.023, -0.004]$, difference $+0.014$ $[+0.001, +0.028]$. The difference is positive on all three wind cells and one of three solar cells, so its effective base is one dataset and it is descriptive.

The footing block's second backbone. Table 1 is the linear mixer because it is dropout-free. The MLP mixer half was run and is reported here rather than withheld, marked conditional on its thread configuration: toggle $+0.029$ $[+0.009, +0.058]$ at global and -0.028 $[-0.053, -0.005]$ at scale-matched, interaction $+0.057$ $[+0.015, +0.107]$ with $6/6$ cells positive, of which the RevIN arm's own footing sensitivity is $+0.024$ $[+0.007, +0.048]$, $6/6$. The boundary also holds on every cell here: the best endogenous configuration across footings trails the parity block's CN by $+0.025$ to $+0.128$, a smaller minimum margin than the linear mixer's $+0.051$ to $+0.147$, which is the margin §3 reports as covariate-conditioned level handling winning every cell. The identity $s\beta(g/s) = \beta g$ is exact only for a linear map, so replication on a nonlinear backbone is evidence the mechanism is not an artifact of the linear case.

Interval width. CN versus RevIN at $h{=}24$ after identical split-conformal recalibration, mean 80% interval width in global-z units with coverage in brackets: GEFCom-Wind 1.12 $[0.82]$ vs 2.22 $[0.76]$; Jeju 1.40 $[0.82]$ vs 1.83 $[0.82]$; GEFCom-Solar 0.44 $[0.82]$ vs 0.69 $[0.80]$; GEFCom-Load 0.54 $[0.79]$ vs 1.07 $[0.82]$. This metric was added after the fact to an existing block and is not itself recorded in advance.

Feeding the statistics back. Also at the GLOBAL footing. If the mechanism is that f never sees $\bar{y}$ or s, the two-line fix is to pass them as extra input channels. On the seven clean cells this closes 6.5% $[1.5, 13.0]$ of the RevIN-to-CN gap (equal-weighted mean of per-cell closure shares) against a withdrawal threshold of 50% recorded in advance but not timestamped (Table 3); the extra channels do help slightly (-0.0115 on the RevIN arm, -0.0047 on the RAW control, so they are not merely harmful), and the largest closure is at $h{=}336$, where the frozen statistic is most stale. The endpoint and its withdrawal clause were fixed after one smoke seed, which is recorded.

The ramp split. Like the mean-only ablation, this decomposes the parity block and therefore inherits its GLOBAL footing; we did not repeat it at another footing. No model was trained for it: the per-origin losses have been on disk since the parity block. Recorded in advance of any array being opened (Table 3), the statistic is $\mathrm{ramp}(t) = \max_k |y_{t+k} - y_{t+k-1}|$ over the horizon, on the realized target only—chosen so that it never references the lookback window, the window mean or a model output, which a definition like $|\bar{y}_{\text{horizon}} - \bar{y}_{\text{lookback}}|$ would and would make the test circular. Equal-count terciles by rank within each cell, ties broken by origin index; value-cut terciles leave an empty bin on the solar set, where many origins are all-night with ramp exactly zero. Bin membership therefore never depends on an arm. As a share of each bin's own error the concentration is 8.0% in the bottom tercile against 9.4% in the top with the interval spanning zero, because high-ramp origins are harder for every arm; we lead with the absolute reading because reserve and ramping are procured in absolute units. On the linear mixer the mean per-cell cost over the seven clean cells is 0.0139 in the bottom tercile and 0.0335 in the top — a $2.4\times$ absolute concentration. The solar cells contribute near-zero or negative costs throughout. Per backbone, top minus bottom over the seven clean cells: linear mixer $+0.0196$ $[+0.0044, +0.0332]$, SegRNN-Cov $+0.0139$ $[+0.0041, +0.0251]$, LightGBM-Cov $+0.0144$ $[-0.0021, +0.0333]$, PatchTST-Cov $+0.0071$ $[-0.0394, +0.0500]$, MLP mixer -0.0195 $[-0.0450, +0.0031]$, its reversal carried entirely by its three wind cells. The recorded secondary that the three terciles be *monotone* fails ($1/7$ cells); only the extremes order. Figure 1(b) uses the lookback each cell's audited models use; the top-versus-bottom ordering also holds at fixed $L \in \{96, 192, 336\}$. At $L{=}720$ it reverses, but that row loses two of the three GEFCom-Wind cells—too few segment-aware windows survive—and on the same reduced base the audited lookback is flat rather than reversed, so we read it as a composition effect, not a counterexample.

A.4 Checks that cross run blocks

Two comparisons that cross run blocks. Both are reported here only. (i) The $11/12$ versus $0/12$ strategy count reads Block E rows (deterministic covariate-conditioned level models and seasonal-naive/climatology anchors at fixed $L{=}336$) against the best instance-normalized arm of the main grid (per-cell tuned lookback). The count reproduces exactly under that reading. Dropping the Load cells removes three of the twelve exogenous strategy-cells, so the honest statement is $9/9$ against an unchanged standard-group base of $0/12$ ($10/12$ and $8/9$ under MASE; every exception is the ridge dynamic regression, on GEFCom-Load under MSE and GEFCom-Solar under MASE), but the two blocks were not run under a common budget, so we report it as indicative rather than as a test. (ii) The capacity-normalized MAE quoted in earlier drafts of §4 was a Block E quantity ($\mathrm{nMAE} = \mathrm{MAE}_{\text{global-}z} \times s^{\mathrm{tr}}/\max^{\mathrm{tr}} y$, Jeju Wind h=24, five-seed means 0.10534 and 0.10005, all five seeds improving), not a cell of the parity block: Block E gives its mixer exogenous *and* calendar covariates where the parity block gives exogenous only. Both arms of that pair see identical covariates, which is the property the claim needs, but the two blocks' MSEs are not interchangeable. Earlier drafts converted this $\Delta\mathrm{nMAE}$ into megawatts of reserve; we no longer do, because reserve is sized on an upper quantile rather than on mean error, and the width delta that would license the conversion is not measured here.

Why the low-LPS side fails, and why regularizing it is not enough. The ETTh first stage does not diverge. Its LightGBM level cannot leave the training range of the target, and none of the 20,160 test values on either dataset does (maximum excursion 0.00 train-y standard deviations). It overfits the calendar: level R^2 falls from 0.916/0.949 in sample to $-0.268/-0.064$ on test (ETTh1/ETTh2, pooled on the global-z scale, the same scale as the reported MSE; per-channel averaging is more negative still, inflated by channels with a near-constant test segment), and $\mathrm{corr}(\hat{\ell}, y)$ collapses from $+0.958/+0.975$ to $+0.278/+0.184$. The two fail by different routes—ETTh1 variance-dominated (test level MSE 1.408, variance 1.015 of it; $\mathrm{sd}(\hat{\ell})/\mathrm{sd}(y)$ rises from 0.920 to 1.158), ETTh2 bias-dominated (test level MSE 3.339, bias^2 2.843 of it, driven by a 1.30 train-sd level shift a fixed calendar function cannot see). Measured at the level layer alone, with no backbone involved, the restored level's test error is $2.13\times/13.78\times$ the RevIN 96-window mean's on ETTh1/ETTh2, against $0.15\times/0.35\times$ on GEFCom-Wind and Jeju Wind. A positive LPS is not a promise that the pointwise level generalizes: on Weather, $\mathrm{LPS} = +0.110$ while the train-fitted level's pointwise test R^2 is -0.015 (different estimands—LPS scores window *means* under expanding-window CV).

A pre-registered block then re-trains CN with the level shrunk toward the train mean, $\tilde{\ell} = \mu + \alpha(\hat{g} - \mu)$, on matched cells (its own tally, never compared with Table 2; the pre-registered rule is a seed mean within each (backbone, h) cell and equal weight over the six cells, which matters—pooling all eighteen rows equally instead gives 3.26/11.48 for the α=1 column rather than 2.75/8.78). The loss-minimizing α tracks LPS: at $\mathrm{LPS} < 0$ the optimum is α=0 (ETTh1 0.386 at α=0 vs 0.387 at α=clip(LPS) and 2.751 at α=1; ETTh2 0.445/0.451/8.781); at $\mathrm{LPS} = 0.11$ and 0.28 it is α=clip(LPS) (Weather 0.177 vs 0.185; Electricity 0.139 vs 0.144); at $\mathrm{LPS} \approx 0.74$ it is α=1. Shrinkage therefore removes the catastrophic tail without ever beating global scaling on the low-LPS side, and *sizing* it requires the quantity LPS measures. The diagnostic-free textbook alternative—fitting α on validation level MSE—is worse than RAW on all four standard datasets, because it minimizes level error rather than forecast error.

Probabilistic transfer and recalibration. CN has the best pinball loss on all four exogenous datasets, the basis of the §4 claim that the boundary transfers. The recalibration cited there is per-quantile additive split conformal [19, 22]: for each level q_k, the q_k-quantile of validation residuals is added to the predicted quantile, followed by re-sorting. Offsets are fitted on the validation segment (which also early-stops the backbone; the test segment is untouched) and applied identically to every arm. On the linear quantile backbone this restores CN's 80% coverage from 0.50–0.66 (per dataset; under-coverage is expected, since the quantile model does not propagate first-stage level uncertainty) to 0.79–0.83 (mean over the four dataset means $0.598 \rightarrow 0.811$; the cell-weighted mean is $0.601 \rightarrow 0.810$), with validation coverage 0.800 by construction; it also *improves* CN's mean pinball over the eleven cells ($0.106 \rightarrow 0.103$; dataset-weighted $0.111 \rightarrow 0.108$), and CN remains the best-pinball arm in all eleven exogenous (dataset, h) cells after every arm is recalibrated. RevIN, already calibrated (0.82), is essentially unchanged, and lead-specific offsets add nothing over pooled ones (0.810 vs 0.811 mean coverage). A central interval can hide tail asymmetry, and here it does: the recalibrated one-sided coverages average 0.101 and 0.912 against nominal 0.10 and 0.90, but GEFCom-Wind sits at 0.147/0.958 and Jeju Wind at 0.054/0.882. Reserve is an upper-tail decision, so this matters; the asymmetry is shared with RevIN (0.143/0.952 on GEFCom-Wind) and is therefore not a property of CN. The LightGBM-Q cross-check agrees ($h \leq 48$; longer-horizon quantile DMS is computationally prohibitive): CN coverage $0.59–0.71 \rightarrow 0.80–0.83$, best pinball in all five cells.

A.5 The score

LPS protocol. With $(\bar{y}_i, \bar{x}_i)$ the target and covariate means of the i-th non-overlapping length-w window and $\hat{g}_k$ the first stage fitted on the windows preceding evaluation block $\mathcal{T}_k$,

$$\text{LPS} \;=\; 1 \;-\; \Big(\sum_k \sum_{i\in\mathcal{T}_k} (\bar{y}_i - \hat{g}_k(\bar{x}_i))^2\Big) \Big/ \Big(\sum_k \sum_{i\in\mathcal{T}_k} (\bar{y}_i - \bar{y}_k^{\text{tr}})^2\Big), \tag{2}$$

with $\bar{y}_k^{\text{tr}}$ the training mean for that block. $w = 96$; first stage LightGBM under expanding chronological cross-validation, fit only on windows preceding each evaluation block; $\tau = 0.3$ was fixed at pre-registration together with the per-dataset sign predictions.

Two variants of the evaluation blocks, one of them not yet run. In the FULL variant — the pre-registered one, and the source of every LPS value printed elsewhere in this paper — the blocks are five equal splits of the final 60% of the non-overlapping windows, and therefore span the whole series, including the segment the grid later tests on. The screen consumes raw window means only, never model outputs or split boundaries, and it is computed before any forecaster is trained; but before training is not before the forecast origin, and the latter is the reading a "pre-training screen" invites. The DEPLOYMENT variant is the identical protocol — same w, same five expanding folds over the final 60% of the windows that exist there — run on the rows preceding the forecast origin alone, that origin being the first row of the grid's own test segment; it is a re-run on the pre-origin prefix rather than a truncation of the frozen fold boundaries, which would change the estimator as well as the data. On Jeju Wind and GEFCom-Wind the two coincide by construction, no recomputation involved: LPS is computed on each series' longest contiguous segment, and that segment ends before the dataset's test segment begins — 2023-06-25 against a test start of 2023-07-04 on Jeju, across the KMA archive hole; 2013-04-22 against 2013-07-10 on GEFCom-Wind, whose curated frame has 42 segments. The remaining six datasets and all ten zones carry a real recomputation, each retaining at least 120 pre-origin windows, well above the six the five folds need.

Result. Every side of τ is retained: $8/8$ datasets and $10/10$ zones, so the FULL variant's sign calls are not an artifact of blocks that reach into the test segment. The values are another matter. On the energy group they barely move (range 0.744–$0.894 \to 0.744$–0.883; Jeju Wind and GEFCom-Wind identical by construction, GEFCom-Load -0.011, GEFCom-Solar -0.026). On the benchmark group they move a great deal: Weather $+0.110 \to -1.347$, ETTh1 $-0.717 \to -0.186$, ETTh2 $-0.205 \to +0.188$, Electricity $0.283 \to 0.2997$. Electricity is the cell to note — it now sits 0.0003 below τ rather than 0.017 below, so the classification survives on a margin three ten-thousandths wide, and the pre-registration's own flag on that dataset as its least confident sign call is vindicated more sharply than we expected. The zones stay above τ throughout (0.471–0.750 against 0.575–0.759), the largest single move being z09 at -0.161. τ was not re-tuned against this recomputation and the committed sign predictions were not re-issued; how each outcome would be read was fixed in writing and committed before the variant was run. Per-series values: `results/lps_deployment.csv` and `results/lps_deployment_zones.csv`.

Remaining LPS **protocol details.** First-stage hyperparameters are fixed (300 trees, learning rate 0.05, 31 leaves). For the multivariate benchmarks LPS is Eq. 2 computed per channel and averaged unweighted over channels. Covariate sets respect the target-time availability requirement (§2) with one exception: the exogenous group uses its forecast covariates plus calendar features, the standard group calendar features only, but GEFCom-Load's is realized temperature, so its LPS of 0.894 — the panel's highest — is optimistic and inflates the bimodality (the body carries this, §2). That is a further reason to read the eight sign hits as two. Co-recorded sibling channels are never used, for LPS or for CN's first stage. The main-grid pre-registration commit precedes the first grid run by four minutes; the zone study's per-zone LPS values and predictions were likewise committed before any zone run. Both are timestamped in the repository below, and Table 3 in §A.6 grades every block on the same three questions.

A.6 Provenance, integrity and availability

Figure 1 construction. One panel, GEFCom-Wind test segment, L=96, h=24. The origin is selected by a stated rule rather than by eye: $\arg\max_t \min\big(|\bar{y}_{[t,t+h)} - \bar{y}_{[t-L,t)}|, |\bar{y}_{[t,t+h)} - y_{t-1}|\big)$, i.e. the test-segment origin whose realized level is furthest from both the window mean and the last observation, so that neither instance normalization's restored level nor persistence anticipates

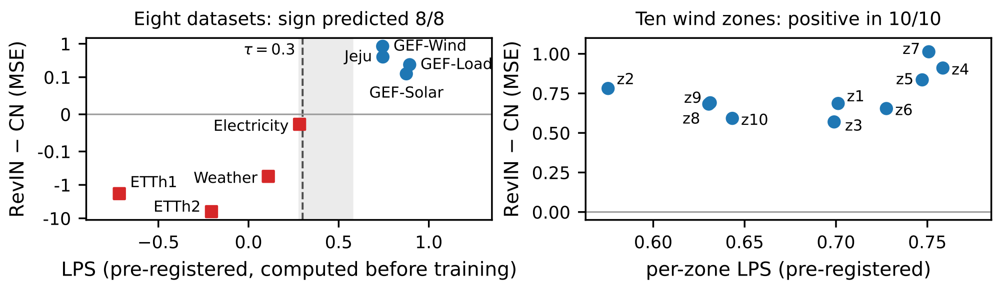

Figure 4: Pre-registered LPS vs realized RevIN−CN gap—the pre-registration receipt for §3. Left: eight datasets (symlog y; dashed τ=0.3; shaded = the 0.283–0.575 stretch between Electricity and the lowest zone, which no dataset or zone occupies — not the whole of $[0.30, 0.70]$, five zones of which lie inside it, §2), all on the predicted side. Right: ten zones, positive in all; ordering inconclusive (n=10, p=0.17).

it. The plotted covariate-implied level is CN's first stage, fitted on train rows only and evaluated on the archived target-time NWP covariates. The panel is an illustration of the mechanism, not a performance claim; the whole-test-segment averages in the caption (0.178 vs 0.042 mean absolute level error over 3,315 origins) are the aggregate version of it. At the selected origin itself RevIN restores 0.671 against a realized 0.150 and an NWP-implied 0.231 — the three values the caption of Fig. 1 used to carry.

Provenance of the claims we make about other people's code. Verified against primary sources rather than recalled. CrossLinear: arXiv:2505.23116v1 is the version consulted (the ACM camera-ready was not reachable); Eqs. 3–4 normalize the endogenous *and* exogenous channels and only the de-normalization uses the target's statistics (Eq. 6); in `models/CrossLinear.py` this is inline parameter-free arithmetic, not an instantiated module, and although `run.py` exposes `--use_norm` the model never reads it and no released script sets it. Its ablations are embedding fusion and positional embedding. TimeXer: the camera-ready contains no occurrence of *normaliz/standardiz/RevIN* outside LayerNorm in the attention equations and the Non-stationary Transformer as a baseline; `--use_norm` defaults to 1 and none of the 17 released scripts sets it to 0. Berthelier et al. [9]: the word *exogenous* occurs once, in a scope sentence; no experimental arm uses covariates.

What our negative claim does not cover. We say only that we are not aware of a table isolating the layer with covariates held identical. Per-dataset sign reversal for instance normalization is itself documented in the covariate-free literature, and TiDE [23] tunes and publishes a per-dataset `revIn` flag without reporting paired on/off numbers; conversely the covariate-aware forecasters fix normalization and toggle covariates instead. We also set aside the classical electricity-price literature on variance-stabilizing transformations, which compares calibration-window transforms of the target rather than per-instance reversible normalization; we have not read it in full.

How much each block's pre-registration is worth. The word is not uniform across this paper, so Table 3 grades every block that carries its own written record on the same three questions. Only the top rows carry a commit timestamp ahead of the run they govern; the body reserves *pre-registered* for those.

Availability. An anonymized repository accompanies this submission — `https://anonymous.4open.science/r/norm-boundary-ccai-B25B` — holding the written pre-registration record for every block of Table 3 in `evidence/`, the frozen result CSVs, the curation and analysis code, and the LPS implementation. `make verify-numbers` re-derives every number this paper prints from those CSVs alone, with no GPU and no re-training. The commit history that timestamps the main-grid and zone-study registrations is preserved in the repository, though the anonymizing proxy serves a file snapshot rather than that history. Raw KMA and GEFCom source data are not redistributed (licensing); collectors and preparation instructions are included.

Table 3: **What "pre-registered" is worth, block by block.** The word is not uniform across this paper and we stop using it as if it were. Only the rows marked *yes* in column 2 carry a commit timestamp ahead of the run they govern; for those the body says *pre-registered*, and for every other block it says *recorded in advance of the run (not timestamped)*. Each record is a file in the accompanying repository, but only the *yes* rows are also commits that predate their run; none was written after the block's own numbers were seen, and wherever column 2 says no, that is our word rather than a timestamp.

Block	Commit timestamped ahead of its run?	Written relative to earlier runs	Amendment
Main grid (§3)	Yes — 4 minutes before the first grid run	Before any forecasting run of this study	None
Zone study	Yes — before the first zone run	After the main grid	None; a deferred LightGBM arm was incomplete and is excluded, disclosed at analysis time
Footing (Table 1)	No	After the main grid had run; the earlier window-normalized block was already run and its result is disclosed inside the record	Yes — same day, after one smoke run and before the grid: the primary endpoint was replaced by a strictly harder one, and both are reported
Mean-only	No	After the main grid and the parity block	None
Fed-back statistics	No	After the parity block	Endpoint and its 50% withdrawal threshold fixed after one smoke seed, which is recorded
Ramp split	No	The per-origin losses were already on disk from the parity block; statistic, bins and endpoints were fixed before any array was opened	None
Shrinkage valve	Yes — committed before either run	After the main grid	Recorded as a post-hoc block, explicitly not part of the original registration
Deployment LPS	Committed, and the run has not happened yet — so the record cannot be back-dated to it	After every block above	None

B The stylized crossover behind τ

Model (M1, linear–Gaussian). One forecast instance: the target-time level is $\ell_T = \Delta + g_T + v_T$, with covariate-explained component $g_T \sim \mathcal{N}(0, \lambda V)$, unexplained component $v_T \sim \mathcal{N}(0, (1-\lambda)V)$, and a covariate-orthogonal train→test shift $\Delta \sim \mathcal{N}(0, \sigma_\Delta^2)$ common to window and target. The lookback window sees stale components $g_w = g_T - \delta_x$ and $v_w = v_T - \delta_u$, where $\delta_x \sim \mathcal{N}(0, s_x^2(h))$ is the within-horizon covariate-driven level change (the ramp) and $\delta_u \sim \mathcal{N}(0, h\sigma_u^2)$ is random-walk drift; the window mean adds shape noise $\bar{\varepsilon} \sim \mathcal{N}(0, \sigma_z^2/w)$. CN's first stage observes $\hat{m} = g_T + e$ with $e \sim \mathcal{N}(0, \sigma_{\text{est}}^2)$. All components are independent.

Risks and crossover. Direct variance accounting of the level-restoration error (a shape residual σ_ε^2 is common to all arms and drops out of comparisons) gives

$$\text{MSE}_{\text{IN}} = s_x^2(h) + h\sigma_u^2 + \sigma_z^2/w, \qquad \text{MSE}_{\text{CN}} = (1-\lambda)V + \sigma_\Delta^2 + \sigma_{\text{est}}^2 :$$

IN pays the staleness of its frozen statistic, CN pays the unexplained level variance. Setting the two equal and solving for λ gives the crossover of §2,

$$\lambda^\star \;=\; 1 \;-\; \frac{s_x^2(h) + h\sigma_u^2 + \sigma_z^2/w - \sigma_\Delta^2 - \sigma_{\text{est}}^2}{V},$$

and $\partial_h\big[\text{MSE}_{\text{IN}} - \text{MSE}_{\text{CN}}\big] = \sigma_u^2 + \partial_h s_x^2 > 0$ (with s_x^2 nondecreasing in h), so the gap grows and $\lambda^\star$ falls with the horizon.

Numerical anchor. The anchor is measured on the validation DGP of the 2,640-run synthetic grid (Fig. 5), not assumed. The DGP sets $m_t = \sqrt{\lambda}\, x_t^\ell + \sqrt{1-\lambda}\, u_t^\ell$, where x^ℓ is the level of an AR(1) covariate (a=0.9), u^ℓ is a standardized random walk, and both are rescaled so that their w=96 window means have unit variance — hence V=1, and λ tracks the level R^2 (matched to within 0.08 in the released tests; the window mean retains a shape component, so the identity is approximate). The covariate-driven ramp therefore scales with λ and the random-walk drift with $1-\lambda$, each living in its own component of the level split; this is why $\lambda^\star$ does not collapse at long horizons — the drift a high-λ series suffers is small by construction. No separate covariate-orthogonal shift is injected

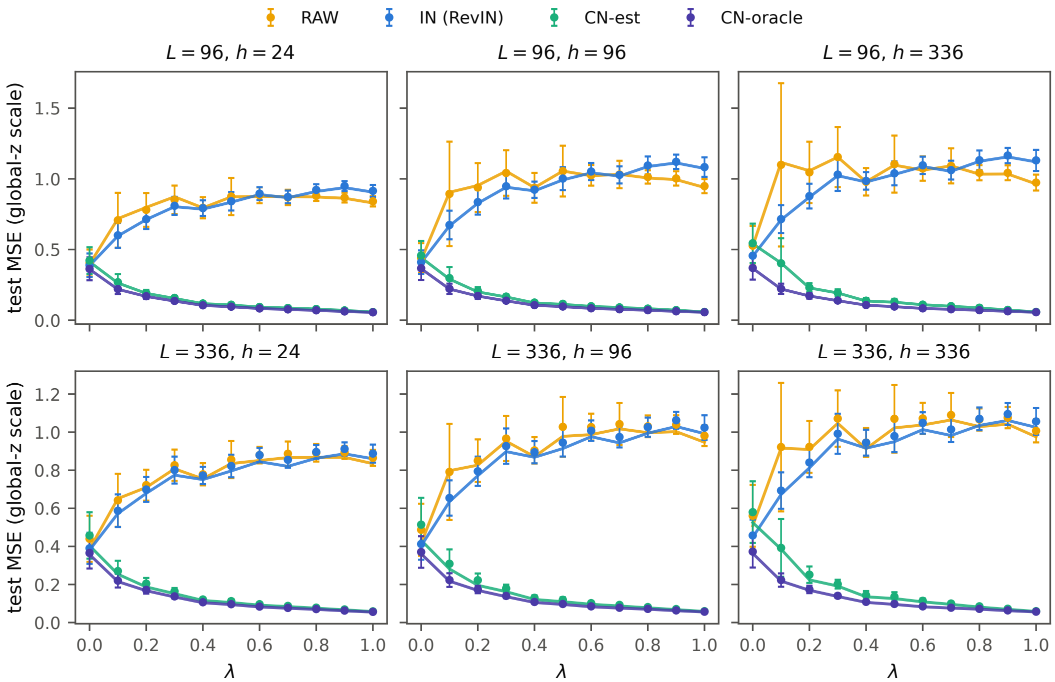

Figure 5: **The synthetic grid behind the anchor.** Seed-mean test MSE (± 1 s.d., ten seeds) against the level-predictability share λ for RAW, RevIN, CN-est and CN-oracle, over $h \in \{24, 96, 336\}$ and $L \in \{96, 336\}$, overlaid on the closed-form constrained-OLS curves of the released test suite; the empirical IN/CN crossings track the closed forms, and the high-λ gap widens with h.

($\sigma^2_\Delta = 0$, a CN-favorable choice, since IN is immune to such shifts), and σ^2_{est} is realized by an actual fitted first stage rather than set. On this DGP the measured restoration errors — $\mathbb{E}[(m_T - \bar{y}_w)^2]$ for IN versus $\mathbb{E}[(m_T - \hat{m}_T)^2]$ for CN — cross at $\lambda^\star \approx 0.3$ at h=24 — the construction is validated only to within 0.08, so the second decimal is not meaningful — declining slightly at h=336 (exact values in the released artifacts): the anchor of §2. The closed forms above are pinned against Monte Carlo at $<1\%$ relative error in the released test suite.

Two conservatism notes. First, the restoration-rule crossover is an *upper bound*: trained linear predictors on the same synthetic grid cross far lower ($\lambda^\star_{\text{OLS}} \approx 0.007$–$0.022$ in closed form, 0.009–0.029 as measured), because backbones implicitly track residual level. A *lower* crossover favours CN—it widens the region in which the endogenous default loses—so this note bounds the anchor conservatively rather than retracting it. That value is a floor under CN-favorable conditions, not a portable constant: $\lambda^\star$ is dataset-specific and rises with the covariate-orthogonal shift σ^2_Δ (zero here), which is largest exactly where instance normalization earns its keep. Second, neither number transports to the LPS axis: LPS is a proxy on a different scale from λ — it absorbs first-stage estimation error and window-mean smoothing, so it is not a consistent estimator of λ, and crossovers computed on the λ scale cannot be read off Fig. 4. τ was fixed on the LPS scale at pre-registration, with the M1 bound anchoring its order of magnitude. The two dominant gaps err in the same direction — the M1 bound overstates the crossover, and first-stage estimation error biases LPS downward relative to λ, both toward instance normalization — while the one choice that cuts the other way (σ^2_Δ=0) shifts the bound by σ^2_Δ/V, a quantity this study does not measure. Conservatism toward instance normalization is the direction the one-sided rule requires: a dataset misclassified below τ forgoes a gain rather than risking the ETTh2-style failure.